NovaTech Industries

Checkpoint 2.4 - Supplier Evaluation Data Dictionary

Document ID: NTI-DATA-002

Version: 1.0 (proposed design)

Status: Draft - pending business approval

Related requirements: FR-005, BR-001 to BR-006, DV-001 to DV-010

Owner: Business Transformation & Analytics Team

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> **Case-study notice:** NovaTech Industries, its suppliers, transactions, and operational figures are fictional. This document specifies a synthetic training environment, not a production system.

1. Purpose and scope

This data dictionary specifies the minimum relational data model required to evaluate supplier quotations, enforce mandatory eligibility rules, calculate weighted scores, and preserve an auditable recommendation. It is the design baseline for synthetic CSV generation and later PostgreSQL implementation. The wider procurement, inventory, and logistics schema will be designed in subsequent checkpoints.

In scope: suppliers, products, supplier capabilities, requisitions, quotations, quotation lines, historical supplier performance, exchange rates, scoring policies, evaluation runs, and per-quotation results.

Out of scope for this checkpoint: purchase-order execution, invoice matching, inventory movement posting, shipment event history, user authentication, and AI agent orchestration.

2. Core data concepts

- **Entity/table:** a collection of records about one business concept, such as suppliers.
- **Row/record:** one instance, such as supplier SUP-0001.
- **Column/field:** one property, such as supplier_status.
- **Primary key (PK):** a unique identifier for a row.
- **Foreign key (FK):** a reference to another table's primary key.
- **Cardinality:** how many records can relate to each other (one-to-many or many-to-many).
- **Nullable:** whether a field may be absent; a missing value is not the same as zero.
- **Snapshot:** a stored copy of values used in a decision, retained so historical evaluations remain reproducible.

3. Data model and relationships

Parent entity	Child entity	Relationship	Purpose
suppliers	supplier_products	1-to-many	Approved products and supply capabilities
products	supplier_products	1-to-many	Product can be offered by multiple suppliers
purchase_requisitions	requisition_items	1-to-many	A requisition may request several products
suppliers	quotations	1-to-many	Supplier can issue many quotations

Parent entity	Child entity	Relationship	Purpose
quotations	quotation_items	1-to-many	Each quotation has one or more priced lines
products	quotation_items	1-to-many	Each quotation line references a product
suppliers	supplier_performance	1-to-many	Period-based performance history
scoring_policies	evaluation_runs	1-to-many	Each run uses one policy version
requisition_items	evaluation_runs	1-to-many	One requisition line can be evaluated repeatedly
evaluation_runs	evaluation_results	1-to-many	One result per assessed quotation line
quotation_items	evaluation_results	1-to-many	Quotation line can appear in later evaluation runs

Design rule: one evaluation run compares quotations for **one requisition item**, one product, one requested quantity, and one required delivery date. A multi-line requisition therefore produces separate evaluation runs for each line. A later optimization phase may consider bundled purchasing across lines.

4. Field-level data dictionary

4.1 suppliers

Grain: one row per supplier. **Primary key:** supplier_id.

Field	Type	Required	Rules / meaning
supplier_id	VARCHAR(20)	Yes	PK; format SUP-0001
supplier_name	VARCHAR(150)	Yes	Nonblank display name
country_code	CHAR(2)	Yes	ISO 3166-1 alpha-2 code
supplier_status	VARCHAR(20)	Yes	active, inactive, suspended
created_at	TIMESTAMP	Yes	Record creation timestamp in UTC

4.2 products

Grain: one row per catalog product. **Primary key:** product_id.

Field	Type	Required	Rules / meaning
product_id	VARCHAR(20)	Yes	PK; format PRD-0001
sku	VARCHAR(40)	Yes	Unique stock-keeping unit
product_name	VARCHAR(150)	Yes	Nonblank description
category_code	VARCHAR(30)	Yes	Controlled product category
unit_of_measure	VARCHAR(12)	Yes	Example: unit, box, kg
product_status	VARCHAR(20)	Yes	active or discontinued

4.3 supplier_products

Grain: one supplier-product capability record. **Primary key:** (supplier_id, product_id).

Field	Type	Required	Rules / meaning
supplier_id	VARCHAR(20)	Yes	PK part; FK -> suppliers
product_id	VARCHAR(20)	Yes	PK part; FK -> products
category_approved	BOOLEAN	Yes	Supplier approved for this product/category
minimum_order_qty	INTEGER	Yes	Greater than zero
maximum_supply_qty	INTEGER	Yes	Greater than or equal to minimum_order_qty
compliance_status	VARCHAR(20)	Yes	approved, pending, rejected
capability_updated_at	TIMESTAMP	Yes	Last capability confirmation, UTC

4.4 purchase_requisitions

Grain: one purchasing request header. **Primary key:** requisition_id.

Field	Type	Required	Rules / meaning
requisition_id	VARCHAR(20)	Yes	PK; format REQ-0001
requested_by_department	VARCHAR(60)	Yes	Example: Operations
requested_at	TIMESTAMP	Yes	Creation timestamp, UTC
requisition_status	VARCHAR(20)	Yes	draft, submitted, approved, cancelled

4.5 requisition_items

Grain: one requested product line. **Primary key:** requisition_item_id.

Field	Type	Required	Rules / meaning
requisition_item_id	VARCHAR(25)	Yes	PK; format REQITEM-0001
requisition_id	VARCHAR(20)	Yes	FK -> purchase_requisitions
product_id	VARCHAR(20)	Yes	FK -> products
requested_quantity	INTEGER	Yes	Greater than zero
required_delivery_date	DATE	Yes	Not before evaluation date for new runs
receiving_location_code	VARCHAR(30)	Yes	Warehouse/site destination identifier

4.6 quotations

Grain: one supplier quotation header. **Primary key:** quotation_id.

Field	Type	Required	Rules / meaning
quotation_id	VARCHAR(20)	Yes	PK; format QUO-0001
supplier_id	VARCHAR(20)	Yes	FK -> suppliers
quotation_date	DATE	Yes	Date supplier issued quotation

Field	Type	Required	Rules / meaning
valid_until	DATE	Yes	On or after quotation_date
currency_code	CHAR(3)	Yes	ISO 4217, e.g. EUR or USD
quotation_status	VARCHAR(20)	Yes	received, withdrawn, accepted, expired

4.7 quotation_items

Grain: one product line on a quotation. **Primary key:** quotation_item_id.

Field	Type	Required	Rules / meaning
quotation_item_id	VARCHAR(25)	Yes	PK; format QUOITEM-0001
quotation_id	VARCHAR(20)	Yes	FK -> quotations
product_id	VARCHAR(20)	Yes	FK -> products
quoted_quantity	INTEGER	Yes	Greater than zero
unit_price	DECIMAL(14,4)	Yes	Strictly positive, quotation currency
freight_cost	DECIMAL(14,2)	Yes	Nonnegative, quotation currency
duty_cost	DECIMAL(14,2)	Yes	Nonnegative, quotation currency
other_cost	DECIMAL(14,2)	Yes	Nonnegative, quotation currency
lead_time_days	INTEGER	Yes	At least 1 in initial model
promised_delivery_date	DATE	Yes	Must be consistent with quoted delivery basis

Cost policy: line_total = requested_quantity * unit_price + freight_cost + duty_cost + other_cost for a quotation that can fulfill the requested quantity. All costs are per quotation line and must be consistently allocated; taxes recoverable by the buyer are excluded unless business policy states otherwise.

4.8 supplier_performance

Grain: one supplier, product category, and rolling evaluation period. **Primary key:** performance_id.

Field	Type	Required	Rules / meaning
performance_id	VARCHAR(25)	Yes	PK
supplier_id	VARCHAR(20)	Yes	FK -> suppliers
category_code	VARCHAR(30)	Yes	Product category covered
period_start	DATE	Yes	Beginning of observation period
period_end	DATE	Yes	End of observation period
inspected_units	INTEGER	Yes	Nonnegative
accepted_units	INTEGER	Yes	Between 0 and inspected_units
completed_deliveries	INTEGER	Yes	Nonnegative
on_time_deliveries	INTEGER	Yes	Between 0 and completed_deliveries

Derived metrics: quality_score = 100 * accepted_units / inspected_units; reliability_score = 100 * on_time_deliveries / completed_deliveries. A zero denominator produces an **unavailable** metric, not zero. No overlapping performance windows for the same supplier/category and evaluation snapshot.

4.9 exchange_rates

Grain: one source currency, target currency, and effective date. **Primary key:** (rate_date, from_currency, to_currency).

Field	Type	Required	Rules / meaning
rate_date	DATE	Yes	Rate's effective business date
from_currency	CHAR(3)	Yes	ISO 4217 source currency
to_currency	CHAR(3)	Yes	ISO 4217 target currency; EUR initially
conversion_rate	DECIMAL(18,8)	Yes	Strictly positive; multiply source amount by rate
rate_source	VARCHAR(80)	Yes	Synthetic reference source identifier

4.10 scoring_policies

Grain: one immutable scoring configuration version. **Primary key:** policy_id.

Field	Type	Required	Rules / meaning
policy_id	VARCHAR(25)	Yes	PK; e.g. POLICY-001-V1
cost_weight	DECIMAL(5,4)	Yes	Default 0.4000
delivery_weight	DECIMAL(5,4)	Yes	Default 0.2500
quality_weight	DECIMAL(5,4)	Yes	Default 0.2000
reliability_weight	DECIMAL(5,4)	Yes	Default 0.1500
normalization_version	VARCHAR(30)	Yes	relative-ratio-v1
effective_from	DATE	Yes	Policy start date
policy_status	VARCHAR(20)	Yes	draft, active, retired

Validation: each weight is between 0 and 1, and all four weights sum to exactly 1.0000. Policies referenced by evaluations cannot be edited in place; create a new version.

4.11 evaluation_runs

Grain: one supplier-comparison event for a requisition item. **Primary key:** evaluation_run_id.

Field	Type	Required	Rules / meaning
evaluation_run_id	VARCHAR(25)	Yes	PK; e.g. EVAL-0001
requisition_item_id	VARCHAR(25)	Yes	FK -> requisition_items
policy_id	VARCHAR(25)	Yes	FK -> scoring_policies
evaluated_at	TIMESTAMP	Yes	UTC evaluation timestamp

Field	Type	Required	Rules / meaning
base_currency	CHAR(3)	Yes	EUR for initial release
run_status	VARCHAR(20)	Yes	completed, review_required, no_eligible_offers
evaluated_by	VARCHAR(60)	Yes	system or simulated user identifier

4.12 evaluation_results

Grain: one assessed quotation line within an evaluation run. **Primary key:** evaluation_result_id. Unique (evaluation_run_id, quotation_item_id).

Field	Type	Required	Rules / meaning
evaluation_result_id	VARCHAR(25)	Yes	PK
evaluation_run_id	VARCHAR(25)	Yes	FK -> evaluation_runs
quotation_item_id	VARCHAR(25)	Yes	FK -> quotation_items
is_eligible	BOOLEAN	Yes	Whether all mandatory rules passed
exclusion_reasons	JSON	Yes	Array of reason codes; empty if eligible
total_cost_eur	DECIMAL(14,2)	No	Comparable converted total; nullable if incomplete
delivery_days_snapshot	INTEGER	No	Lead time used for scoring
quality_score	DECIMAL(6,2)	No	0-100, nullable if insufficient history
reliability_score	DECIMAL(6,2)	No	0-100, nullable if insufficient history
cost_score	DECIMAL(6,2)	No	0-100; computed among eligible offers
delivery_score	DECIMAL(6,2)	No	0-100; computed among eligible offers
final_score	DECIMAL(6,2)	No	0-100; null for ineligible/incomplete
recommendation_rank	INTEGER	No	1 is best; null when not ranked
requires_manual_review	BOOLEAN	Yes	True if history or data confidence inadequate

Auditability: preserve the source quotation, performance period, FX effective date, scoring policy version, and eligible comparison set. The production implementation may add an immutable JSON evaluation snapshot to guarantee reproducibility when source records change.

5. Worked example and evaluation lineage

Requisition: REQ-0001, line REQITEM-0001, product PRD-0001, quantity 100, required delivery date 2026-11-20.

Supplier	Comparable cost (EUR)	Lead time	Quality	Reliability
EuroParts GmbH	12,000	5 days	98	96
Shenzhen Components	10,000	12 days	92	84
Nordic Industrial AB	11,500	7 days	99	95

Assuming all three quotations are eligible, the minimum cost is EUR 10,000 and minimum lead time is 5 days. Using policy weights 40/25/20/15:

Supplier	Cost score	Delivery score	Quality	Reliability	Weighted total
EuroParts GmbH	83.33	100.00	98	96	92.33
Shenzhen Components	100.00	41.67	92	84	81.42
Nordic Industrial AB	86.96	71.43	99	95	86.69

Recommendation: EuroParts GmbH. Values are rounded for display; scoring calculations must retain unrounded intermediate precision. The ranking is valid only for this evaluation set and policy version.

6. Validation and integrity controls

Control	Enforcement
Unique IDs and SKU	PK / unique constraints
Valid references	Foreign-key constraints
Valid statuses and currencies	Controlled vocabularies / checks
Positive quantities and prices	Database CHECK constraints and API validation
Valid quotation dates	valid_until >= quotation_date
Valid performance counts	Accepted <= inspected; on-time <= completed
Nonnegative freight/duty/other costs	Database CHECK constraints
Weight sum = 100%	Policy validation before activation
Eligible quotation consistency	Product, quantity, date, compliance, supplier status
Missing historical metrics	Manual review; do not fabricate scores
Currency conversion	Record rate date, rate source, and applied rate
Reproducibility	Immutable policy version and evaluation snapshot

7. Synthetic data generation plan

The first practical exercise will use small CSV files. We will expand to relational tables only after validating the basic pipeline.

Stage	Dataset size	Learning objective
Starter	3 suppliers, 10 products, 3 quotations	CSV reading and simple scoring
Intermediate	20 suppliers, 50 products, 100 quotations	Foreign keys, joins, validation
Advanced	50 suppliers, 200 products, 2,500 quotation headers	Multi-table data generation, history, performance

Generation rules: deterministic random seed; stable unique IDs; referential integrity; plausible country/lead-time relationships; nonnegative prices; realistic quality and delivery histories; explicit missing-data scenarios; separate deliberately-invalid test fixtures; and no real personal or commercial information.

Recommended starter files: suppliers.csv, products.csv, supplier_products.csv, requisition_items.csv, quotations.csv, quotation_items.csv, and supplier_performance.csv. Additional policy and FX tables can be introduced when the exercise reaches multi-currency and historical scoring.

8. Open decisions for business approval

1. **Historical sufficiency:** minimum inspected units and completed deliveries before automatic scoring (proposed initial thresholds: 100 inspected units and 10 completed deliveries).
2. **Delivery basis:** confirm whether supplier lead time is measured from purchase order release or quotation acceptance; use calendar days consistently.
3. **Currency policy:** select FX rate effective on evaluation date, or the latest available rate not after that date.
4. **Tie-breaking:** proposed sequence: higher weighted score, lower total cost, shorter lead time, then stable quotation ID.
5. **Performance granularity:** category-specific historical scores are preferred; define fallback when category-level history is sparse.
6. **Approval authority:** Procurement Officer reviews all recommendations; Procurement Manager reviews missing-history cases and exceptions.

These decisions are **proposals**, not yet approved rules. They must be finalized before production-grade evaluation logic is implemented.

9. Acceptance criteria for the data model

- All evaluation results reference valid quotation lines and evaluation runs.
- A quotation cannot be ranked unless mandatory eligibility rules are satisfied and all required scores are available.
- Scores can be reproduced from saved inputs, policy version, and comparison set.
- Supplier quality and reliability rates derive from recorded numerators and denominators.
- Every foreign key resolves to an existing parent record.
- Synthetic datasets contain no real customer, supplier, or employee information.
- CSV exports can be imported into PostgreSQL without changing field meaning.

10. Repository location and next checkpoint

Save the two versions of this document as:

docs/data/supplier-evaluation-data-dictionary.md
docs/data/supplier-evaluation-data-dictionary.pdf

Next checkpoint 2.5: approve the outstanding data policies, define the starter CSV schema, and generate a small coherent synthetic dataset with Python. No database or Docker installation is required yet.